Chenxu Liu 刘宸旭

B.Sc. in Physics — Shandong University, Qingdao

202422201069@mail.sdu.edu.cn · physicsliu299792458@outlook.com · github.com/PhyLiusir
Qingdao, China
Chinese (native) · English (working proficiency)

EDUCATION

Shandong University, Qingdao

2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Physics for the foundations. Coursework: theoretical mechanics, electrodynamics, thermodynamics and statistical physics, quantum mechanics, advanced quantum mechanics, quantum field theory, particle physics, general relativity. Time outside it goes to engineering on both sides of large models and agents — read, reproduce, then change it on purpose.

HOW I WORK

Derive from first principles, not from precedent — ugly is fine, fabricated is not. Every claim has to replay from a log.

RESEARCH INTERESTS

Large models: deployment and serving economics, post-training (SFT → preference → RL), and new attention architectures — Loop Attention.
Agents: agent construction, tool design, multi-agent collaboration, retrieval-augmented generation.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Deployment (LM 01).** Cost structure of one model across serving stacks, batching and precision — cost per token, P99 latency and peak memory read together.
- **Post-training (LM 02).** A full SFT → preference → RL chain reproduced at small scale, then ablated stage by stage; focus on which gains survive on a smaller model.
- **Loop Attention (LM 03).** Reusing one attention block across depth — memory, wall-clock and training stability measured against a stacked baseline. Main line.
- **Agent construction (AG 01).** The smallest useful combination of loop, memory and planning; a taxonomy of tool-less agent failures as the starting object.
- **Tool design (AG 02).** Schemas, error surfaces and return shapes for a batch of real tasks, comparing failure distributions before and after.
- **Multi-agent collaboration (AG 03).** A strong single-agent baseline first; multi-agent setups have to beat it reliably before roles get split.
- **RAG (AG 04).** A citation-required QA set and a scoring script: an answer must land on a specific passage or it counts as wrong.

CURRENT WORK

- **LM 03 · Loop Attention — the bill for depth reuse.** memory, wall-clock and training stability measured against a stacked baseline.
- **LM 01 · Serving comparison log.** throughput, memory and tail latency for one model across serving and quantisation configs.
- **AG 02 · Tool interface.** schemas and error handling redone for a batch of real tasks, with the failure modes logged.
- **AG 04 · Citation-required RAG eval set.** an answer has to land on a specific passage, or it counts as wrong.

OPEN QUESTIONS

- **LM 01 · Deployment.** At what load does the batching-versus-tail-latency trade-off flip sign? · Is the throughput bought with quantisation worth the accuracy it costs? · Should the KV-cache memory budget be accounted per request or per tenant?
- **LM 02 · Post-training.** Which data-selection metric actually predicts downstream gain? · What observable signals reveal that an eval set has been contaminated? · At what model size do RL-stage gains start to appear?
- **LM 03 · New attention architecture — Loop Attention.** Where is the trade-off point between depth reuse and width scaling? · Which metric detects the representation collapse caused by reuse, early? · On long-context tasks, could reuse turn from cost into advantage?
- **AG 01 · Building agents.** Which failure modes can extra tools fix, and which can they not? · Who should decide when a memory gets written? · At what task length does planning start to pay for itself?
- **AG 02 · Tool design.** How detailed must an error message be before the model actually changes behaviour? · Is it better for tools to be coarse-grained or fine-grained? · How structured should a return value be?
- **AG 03 · Multi-agent collaboration.** How does coordination overhead grow with the number of roles? · What signal says roles should be merged rather than split further? · How does failure propagate between roles?
- **AG 04 · RAG.** In what range is the chunk-size versus recall relationship most stable? · Should a wrong citation cost more than no citation at all? · At what corpus size does re-ranking start to pay off?

TECHNICAL SKILLS

LANGUAGES	Python (primary), C++ and CUDA, Bash, Git and code review
MODEL SIDE	PyTorch training and fine-tuning, transformers and custom layers, serving and quantisation, LoRA and preference alignment
AGENT SIDE	Tools and function calling, multi-agent orchestration, vector retrieval and RAG, evaluation sets and scoring scripts
TOOLING	Linux and containers, distributed training basics, reproducible experiments, technical writing